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Problems | Output | Debug Console | Terminal | bash - lab3-fork-exec

```
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch10_pool 3 file1 file2 file3 file
4 file5
[parent] worker PID=43191 finished
[worker PID=43210] starting task: file4
[worker PID=43189] finished task: file1
[worker PID=43190] finished task: file2
[parent] worker PID=43189 finished
[parent] worker PID=43190 finished
[worker PID=43211] starting task: file5
[worker PID=43210] finished task: file4
[parent] worker PID=43210 finished
[worker PID=43211] finished task: file5
[parent] worker PID=43211 finished
[parent] all tasks completed
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ make clean
rm -rf bin build
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ make
mkdir -p bin
gcc -Wall -Wextra -std=gnu11 -g src/exec_demo.c -o bin/exec_demo
gcc -Wall -Wextra -std=gnu11 -g src/fork_demo.c -o bin/fork_demo
gcc -Wall -Wextra -std=gnu11 -g src/pipe_demo.c -o bin/pipe_demo
gcc -Wall -Wextra -std=gnu11 -g src/pipeline_demo.c -o bin/pipeline_demo
gcc -Wall -Wextra -std=gnu11 -g src/zombie_demo.c -o bin/zombie_demo
gcc -Wall -Wextra -std=gnu11 -g challenges/ch10_pool.c -o bin/ch10_pool
gcc -Wall -Wextra -std=gnu11 -g challenges/ch1_single_fork.c -o bin/ch1_single_fork
gcc -Wall -Wextra -std=gnu11 -g challenges/ch2_multi_child.c -o bin/ch2_multi_child
gcc -Wall -Wextra -std=gnu11 -g challenges/ch3_exec_ls.c -o bin/ch3_exec_ls
gcc -Wall -Wextra -std=gnu11 -g challenges/ch4_exec_worker.c -o bin/ch4_exec_worker
gcc -Wall -Wextra -std=gnu11 -g challenges/ch5_exec_examples.c -o bin/ch5_exec_examples
gcc -Wall -Wextra -std=gnu11 -g challenges/ch6_pipe_sum.c -o bin/ch6_pipe_sum
gcc -Wall -Wextra -std=gnu11 -g challenges/ch7_pipeline.c -o bin/ch7_pipeline
gcc -Wall -Wextra -std=gnu11 -g challenges/ch8_wait_nonblock.c -o bin/ch8_wait_nonblock
gcc -Wall -Wextra -std=gnu11 -g challenges/ch9_zombie.c -o bin/ch9_zombie
gcc -Wall -Wextra -std=gnu11 -g challenges/worker.c -o bin/worker
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ls bin
ch10_pool        ch3_exec_ls        ch6_pipe_sum       ch9_zombie  pipe_demo      zombie_demo
ch1_single_fork  ch4_exec_worker    ch7_pipeline       exec_demo   pipeline_demo
ch2_multi_child  ch5_exec_examples  ch8_wait_nonblock  fork_demo   worker
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./bin/ch4_exec_worker
worker: arguments received:
argv[0] = worker
argv[1] = one
argv[2] = two
MYVAR=hello
[parent] worker finished
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $
```

Chat

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

Tip: Use /create-agent to scaffold custom agent for your workflow. Disable tips

C ch4_exec_worker.c:22

Describe what to build

Agent | Models

Local | Default permissions

New Tab | Lab 3.1: Fork & Exec | (695) Nuclear Historian Re | (HCSC-375-01) Operating | Google Docs: Online Docu | Lab3 - Fork and Exec - Go | ch1_single_fork.c — linux-

urban-space-giggle-97j479545gwjcx55v.github.dev

linux-c-lab [Codespaces: urban space giggle]

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Outline

Timeline

.c U | pipeline_demo.c U | zombie_demo.c U | Makefile U | ch1_single_fork.c U

lab3-fork-exec > challenges > ch1_single_fork.c

```
44  }
```

Problems | Output | Debug Console | Terminal | bash - lab3-fork-exec

```
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ls -la
drwxrwxrwx+ 2 codespace codespace  4096 Sep  9 01:42 challenges
-rwxrwxrwx  1 codespace codespace 17904 Sep  9 01:49 exec_demo
-rwxrwxrwx  1 codespace codespace 18208 Sep  9 01:47 fork_demo
drwxrwxrwx+ 2 codespace codespace  4096 Sep  9 01:42 lab_report
-rwxrwxrwx  1 codespace codespace 19592 Sep  9 01:51 pipe_demo
-rwxrwxrwx  1 codespace codespace 18264 Sep  9 01:56 pipeline_demo
drwxrwxrwx+ 2 codespace codespace  4096 Sep  9 02:00 src
-rwxrwxrwx  1 codespace codespace 18000 Sep  9 01:58 zombie_demo
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ make
mkdir -p bin
gcc -Wall -Wextra -std=gnu11 -g src/exec_demo.c -o bin/exec_demo
gcc -Wall -Wextra -std=gnu11 -g src/fork_demo.c -o bin/fork_demo
gcc -Wall -Wextra -std=gnu11 -g src/pipe_demo.c -o bin/pipe_demo
gcc -Wall -Wextra -std=gnu11 -g src/pipeline_demo.c -o bin/pipeline_demo
gcc -Wall -Wextra -std=gnu11 -g src/zombie_demo.c -o bin/zombie_demo
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ls
Makefile  bin  challenges  exec_demo  fork_demo  lab_report  pipe_demo  pipeline_demo  src  zombie_demo
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ls bin
exec_demo  fork_demo  pipe_demo  pipeline_demo  zombie_demo
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ gcc -Wall -Wextra -std=gnu11 -g chal
lenges/ch1_single_fork.c -o ch1_single_fork
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch1_single_fork
Hello from child (PID=16579, PPID=16578)
child 16579 exited with status 7
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $
```

Chat

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

Tip: Use /create-agent to scaffold a custom agent for your workflow.

C ch1_single_fork.c

Describe what to build

Agent | Models

Local | Default permissions

Codespaces: urban space giggle | lab3-fork-exec* | 0 0 | 0 | Ln 44, Col 2 | Spaces: 4 | UTF-8 | LF | { } C | Sign In | Layout: U.S.

New Tab | Lab 3.1: Fork & Exec | (695) Nuclear Histori | (HCSC-375-01) Oper | Google Docs: Online | Lab3 - Fork and Exec | ch2_multi_child.c — l | how to get a file icon

urban-space-giggle-97j479545gwjcx55v.github.dev

linux-c-lab [Codespaces: urban space giggle]

Explorer

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Outline

Timeline

c U | zombie_demo.c U | Makefile U | ch1_single_fork.c U | ch2_multi_child.c U

lab3-fork-exec > challenges > ch2_multi_child.c

```
6   int main(int argc, char *argv[]) {
39      for (int i = 0; i < n; i++) {
49          if (WIFEXITED(status)) {
55          }
56      }
57
58      return 0;
59  }
```

Problems | Output | Debug Console | Terminal | bash - lab3-fork-exec

```
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ make
gcc -Wall -Wextra -std=gnu11 -g src/pipeline_demo.c -o bin/pipeline_demo
gcc -Wall -Wextra -std=gnu11 -g src/zombie_demo.c -o bin/zombie_demo
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ls
Makefile  bin  challenges  exec_demo  fork_demo  lab_report  pipe_demo  pipeline_demo  src  zombie_demo
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ls bin
exec_demo  fork_demo  pipe_demo  pipeline_demo  zombie_demo
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ gcc -Wall -Wextra -std=gnu11 -g chal
lenges/ch1_single_fork.c -o ch1_single_fork
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch1_single_fork
Hello from child (PID=16579, PPID=16578)
child 16579 exited with status 7
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ gcc -Wall -Wextra -std=gnu11 -g chal
lenges/ch2_multi_child.c -o ch2_multi_child
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch2_multi_child 5
Child index 0: PID=18292, PPID=18291
Child index 1: PID=18293, PPID=18291
Child index 2: PID=18294, PPID=18291
Child index 3: PID=18295, PPID=18291
Parent: child PID=18292 finished with status 1
Parent: child PID=18293 finished with status 2
Parent: child PID=18294 finished with status 3
Parent: child PID=18295 finished with status 4
Child index 4: PID=18296, PPID=18291
Parent: child PID=18296 finished with status 5
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $
```

Chat

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

Tip: Use /create-agent to scaffold a custom agent for your workflow.

C ch2_multi_child.c

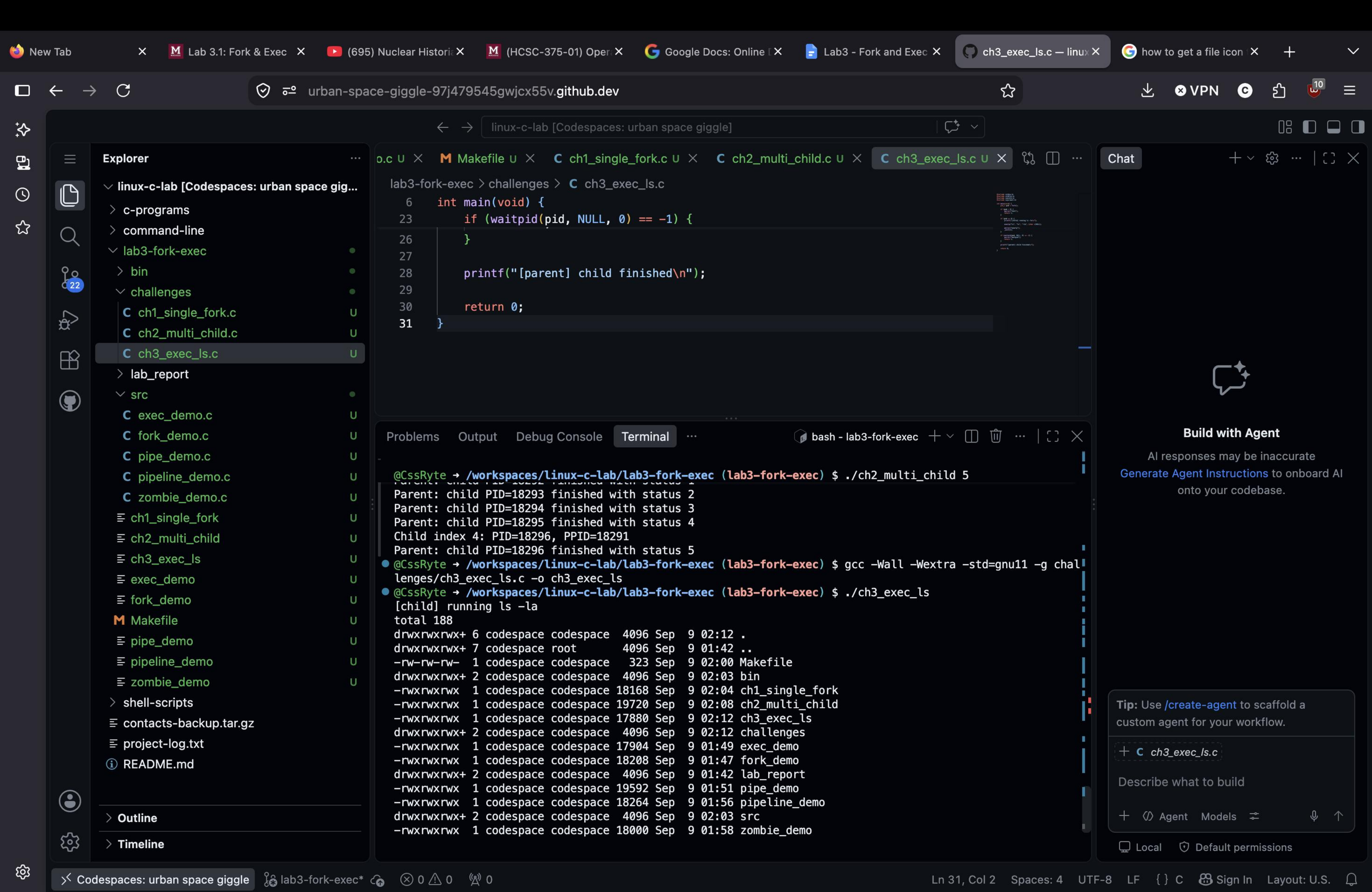
Describe what to build

Agent | Models

cal | Default permissions

Select Encoding

Codespaces: urban space giggle | lab3-fork-exec* | 0 0 | 0 | Ln 59, Col 2 | Spaces: 4 | UTF-8 | LF | { } C | Sign In | Layout: U.S.


New Tab
Lab 3.1: Fork & Exec
(695) Nuclear Histori
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Google Docs: Online
Lab3 - Fork and Exec
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urban-space-giggle-97j479545gwjcx55v.github.dev
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lab3-fork-exec > challenges > ch3_exec_ls.c
6 int main(void) {
23 if (waitpid(pid, NULL, 0) == -1) {
26 }
27
28 printf("[parent] child finished\n");
29
30 return 0;
31 }
Problems
Output
Debug Console
Terminal
bash - lab3-fork-exec
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch2_multi_child 5
Parent: child PID=18293 finished with status 2
Parent: child PID=18294 finished with status 3
Parent: child PID=18295 finished with status 4
Child index 4: PID=18296, PPID=18291
Parent: child PID=18296 finished with status 5
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ gcc -Wall -Wextra -std=gnu11 -g challenges/ch3_exec_ls.c -o ch3_exec_ls
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch3_exec_ls
[child] running ls -la
total 188
drwxrwxrwx+ 6 codespace codespace 4096 Sep 9 02:12 .
drwxrwxrwx+ 7 codespace root 4096 Sep 9 01:42 ..
-rw-rw-rw- 1 codespace codespace 323 Sep 9 02:00 Makefile
drwxrwxrwx+ 2 codespace codespace 4096 Sep 9 02:03 bin
-rwxrwxrwx 1 codespace codespace 18168 Sep 9 02:04 ch1_single_fork
-rwxrwxrwx 1 codespace codespace 19720 Sep 9 02:08 ch2_multi_child
-rwxrwxrwx 1 codespace codespace 17880 Sep 9 02:12 ch3_exec_ls
drwxrwxrwx+ 2 codespace codespace 4096 Sep 9 02:12 challenges
-rwxrwxrwx 1 codespace codespace 17904 Sep 9 01:49 exec_demo
-rwxrwxrwx 1 codespace codespace 18208 Sep 9 01:47 fork_demo
drwxrwxrwx+ 2 codespace codespace 4096 Sep 9 01:42 lab_report
-rwxrwxrwx 1 codespace codespace 19592 Sep 9 01:51 pipe_demo
-rwxrwxrwx 1 codespace codespace 18264 Sep 9 01:56 pipeline_demo
drwxrwxrwx+ 2 codespace codespace 4096 Sep 9 02:03 src
-rwxrwxrwx 1 codespace codespace 18000 Sep 9 01:58 zombie_demo
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Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Tip: Use /create-agent to scaffold a custom agent for your workflow.
ch3_exec_ls.c
Describe what to build
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Outline

Timeline

c U | ch2_multi_child.c U | ch3_exec_ls.c U | worker.c U | ch4_exec_worker.c U

lab3-fork-exec > challenges > ch4_exec_worker.c

```
6   int main(void) {
33      if (waitpid(pid, NULL, 0) == -1) {
36      }
37
38      printf("[parent] worker finished\n");
39
40      return 0;
41  }
```

Problems | Output | Debug Console | Terminal | bash - lab3-fork-exec

```
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch4_exec_worker
worker: arguments received:
argv[0] = worker
argv[1] = one
argv[2] = two
MYVAR=hello
[parent] worker finished
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $
```

Chat

Build with Agent

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Generate Agent Instructions to onboard AI onto your codebase.

Tip: Use /create-agent to scaffold a custom agent for your workflow.

C ch4_exec_worker.c

Describe what to build

Agent | Models

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Explorer

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Outline

Timeline

ch3_exec_ls.c U | worker.c U | ch4_exec_worker.c U | ch5_exec_examples.c U

lab3-fork-exec > challenges > ch5_exec_examples.c

```
6   int main(void) {
36      if (child_b == 0) {
48      }
49
50      waitpid(child_b, NULL, 0);
51
52      return 0;
53  }
```

Problems | Output | Debug Console | Terminal | bash - lab3-fork-exec

```
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch4_exec_worker
worker: arguments received:
argv[0] = worker
argv[1] = one
argv[2] = two
MYVAR=hello
[parent] worker finished
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ gcc -Wall -Wextra -std=gnu11 -g chal
lenges/ch5_exec_examples.c -o ch5_exec_examples
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch5_exec_examples
one two
one two
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $
```

Chat

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

Tip: Use /create-agent to scaffold a custom agent for your workflow.

ch5_exec_examples.c

Describe what to build

Agent | Models

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  - src
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    - fork_demo.c U
    - pipe_demo.c U
    - pipeline_demo.c U
    - zombie_demo.c U
  - ch1_single_fork U
  - ch2_multi_child U
  - ch3_exec_ls U
  - ch4_exec_worker U
  - ch5_exec_examples U
  - ch6_pipe_sum U
  - exec_demo U
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Outline

Timeline

worker.c U | ch4_exec_worker.c U | ch5_exec_examples.c U | ch6_pipe_sum.c U

lab3-fork-exec > challenges > ch6_pipe_sum.c

```
6   int main(void) {
42
43      close(fd[1]);
44
45      waitpid(pid, NULL, 0);
46
47      return 0;
48  }
```

Problems | Output | Debug Console | Terminal | bash - lab3-fork-exec

```
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch4_exec_worker
worker: arguments received:
argv[0] = worker
argv[1] = one
argv[2] = two
MYVAR=hello
[parent] worker finished
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ gcc -Wall -Wextra -std=gnu11 -g chal
lenges/ch5_exec_examples.c -o ch5_exec_examples
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch5_exec_examples
one two
one two
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ gcc -Wall -Wextra -std=gnu11 -g chal
lenges/ch6_pipe_sum.c -o ch6_pipe_sum
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch6_pipe_sum
Sum = 55
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $
```

Chat

Build with Agent

AI responses may be inaccurate

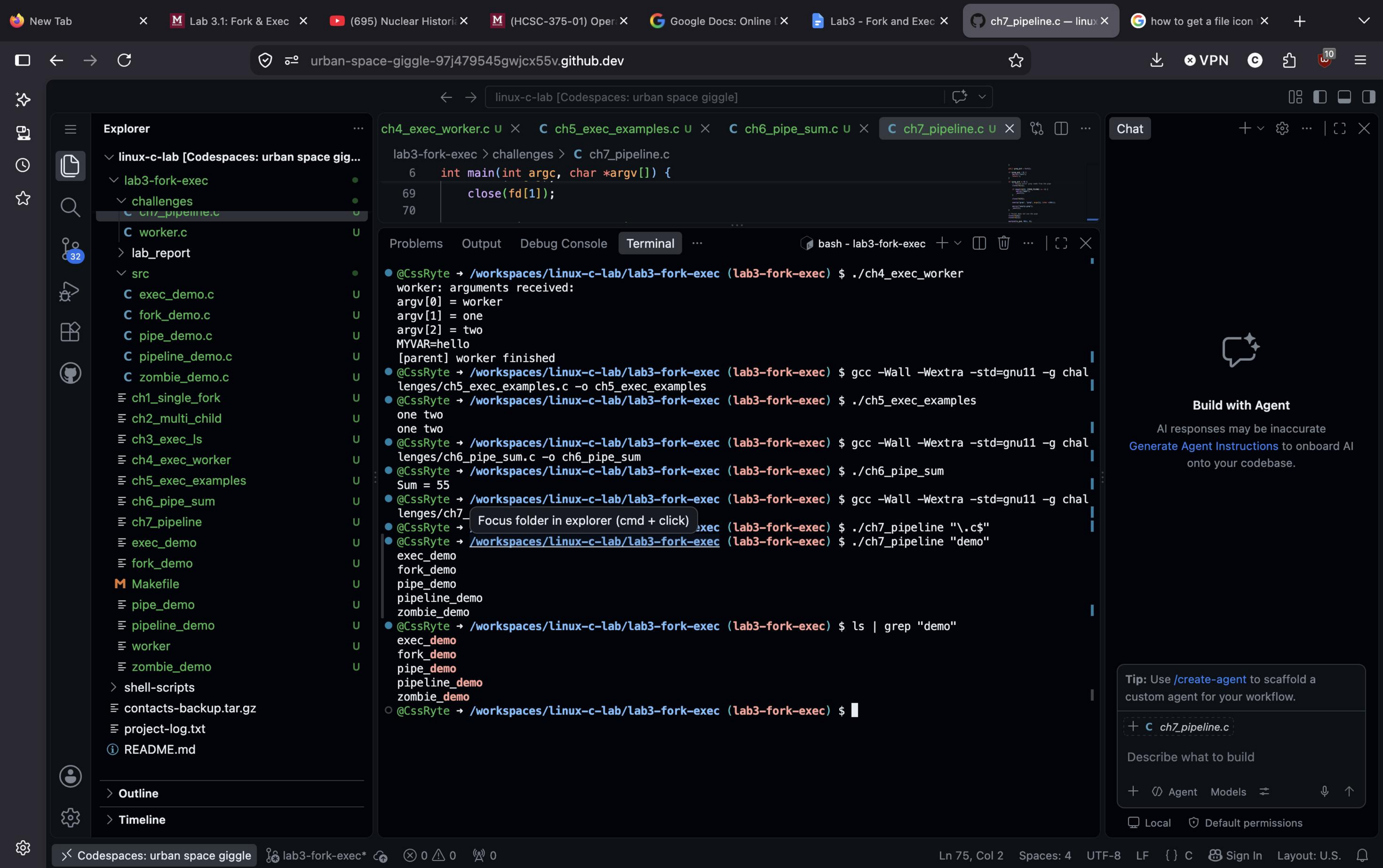
Generate Agent Instructions to onboard AI onto your codebase.

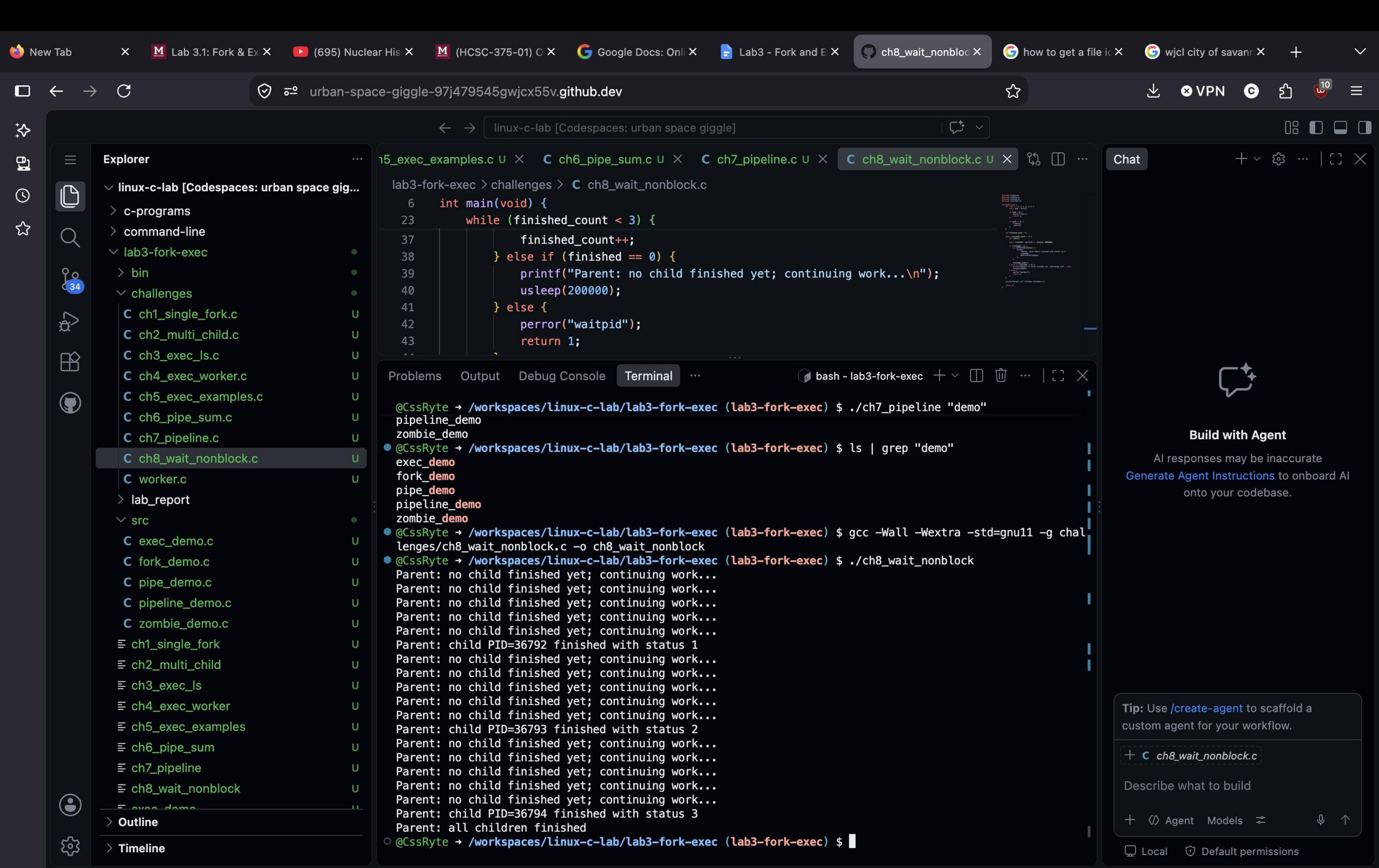
Tip: Use /create-agent to scaffold a custom agent for your workflow.

ch6_pipe_sum.c

Unable to read from the browser's clipboard. Please make sure you have granted access for this website to read from the clipboard.

Retry | Learn More


New Tab
Lab 3.1: Fork & Exec
(695) Nuclear Histori
(HCSC-375-01) Oper
Google Docs: Online
Lab3 - Fork and Exec
ch7_pipeline.c — linux
how to get a file icon
urban-space-giggle-97j479545gwjcx55v.github.dev
VPN
linux-c-lab [Codespaces: urban space giggle]
Explorer
linux-c-lab [Codespaces: urban space gig...
lab3-fork-exec
challenges
worker.c
lab_report
src
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fork_demo.c
pipe_demo.c
pipeline_demo.c
zombie_demo.c
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ch2_multi_child
ch3_exec_ls
ch4_exec_worker
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ch6_pipe_sum
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pipe_demo
pipeline_demo
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project-log.txt
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Timeline
ch4_exec_worker.c U
ch5_exec_examples.c U
ch6_pipe_sum.c U
ch7_pipeline.c U
lab3-fork-exec > challenges > ch7_pipeline.c
6 int main(int argc, char *argv[]) {
69 close(fd[1]);
70
Problems
Output
Debug Console
Terminal
bash - lab3-fork-exec
@CssRyte → /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch4_exec_worker
worker: arguments received:
argv[0] = worker
argv[1] = one
argv[2] = two
MYVAR=hello
[parent] worker finished
@CssRyte → /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ gcc -Wall -Wextra -std=gnu11 -g challenges/ch5_exec_examples.c -o ch5_exec_examples
@CssRyte → /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch5_exec_examples
one two
one two
@CssRyte → /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ gcc -Wall -Wextra -std=gnu11 -g challenges/ch6_pipe_sum.c -o ch6_pipe_sum
@CssRyte → /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch6_pipe_sum
Sum = 55
@CssRyte → /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ gcc -Wall -Wextra -std=gnu11 -g challenges/ch7_
Focus folder in explorer (cmd + click)
@CssRyte → /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch7_pipeline "\.c$"
@CssRyte → /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch7_pipeline "demo"
exec_demo
fork_demo
pipe_demo
pipeline_demo
zombie_demo
@CssRyte → /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ls | grep "demo"
exec_demo
fork_demo
pipe_demo
pipeline_demo
zombie_demo
@CssRyte → /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $
Chat
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Tip: Use /create-agent to scaffold a custom agent for your workflow.
ch7_pipeline.c
Describe what to build
Agent
Models
Local
Default permissions
Codespaces: urban space giggle
lab3-fork-exec*
Ln 75, Col 2
Spaces: 4
UTF-8
LF
C
Sign In
Layout: U.S.


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Lab 3.1: Fork & Ex
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Lab3 - Fork and E
ch8_wait_nonbloc
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urban-space-giggle-97j479545gwjcx55v.github.dev
VPN
linux-c-lab [Codespaces: urban space giggle]
Explorer
linux-c-lab [Codespaces: urban space gig...
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ch3_exec_ls.c U
ch4_exec_worker.c U
ch5_exec_examples.c U
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ch8_wait_nonblock.c U
worker.c U
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src
exec_demo.c U
fork_demo.c U
pipe_demo.c U
pipeline_demo.c U
zombie_demo.c U
ch1_single_fork U
ch2_multi_child U
ch3_exec_ls U
ch4_exec_worker U
ch5_exec_examples U
ch6_pipe_sum U
ch7_pipeline U
ch8_wait_nonblock U
Outline
Timeline
ch5_exec_examples.c U
ch6_pipe_sum.c U
ch7_pipeline.c U
ch8_wait_nonblock.c U
lab3-fork-exec > challenges > ch8_wait_nonblock.c
6 int main(void) {
23 while (finished_count < 3) {
37 finished_count++;
38 } else if (finished == 0) {
39 printf("Parent: no child finished yet; continuing work...\n");
40 usleep(200000);
41 } else {
42 perror("waitpid");
43 return 1;
Problems
Output
Debug Console
Terminal
bash - lab3-fork-exec
@CssRyte → /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch7_pipeline "demo"
pipeline_demo
zombie_demo
@CssRyte → /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ls | grep "demo"
exec_demo
fork_demo
pipe_demo
pipeline_demo
zombie_demo
@CssRyte → /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ gcc -Wall -Wextra -std=gnu11 -g challenges/ch8_wait_nonblock.c -o ch8_wait_nonblock
@CssRyte → /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch8_wait_nonblock
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: child PID=36792 finished with status 1
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: child PID=36793 finished with status 2
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: child PID=36794 finished with status 3
Parent: all children finished
@CssRyte → /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $
Chat
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ch8_wait_nonblock.c
Describe what to build
Agent
Models
Local
Default permissions
Codespaces: urban space giggle
lab3-fork-exec*
0 0
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Ln 50, Col 2
Spaces: 4
UTF-8
LF
C
Sign In
Layout: U.S.

Explorer

linux-c-lab [Codespaces: urban space gig...

- c-programs
- command-line
- lab3-fork-exec
  - bin
  - challenges
    - ch1_single_fork.c U
    - ch2_multi_child.c U
    - ch3_exec_ls.c U
    - ch4_exec_worker.c U
    - ch5_exec_examples.c U
    - ch6_pipe_sum.c U
    - ch7_pipeline.c U
    - ch8_wait_nonblock.c U
    - ch9_zombie.c U
    - worker.c U
  - lab_report
  - src
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    - fork_demo.c U
    - pipe_demo.c U
    - pipeline_demo.c U
    - zombie_demo.c U
  - ch1_single_fork U
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  - ch3_exec_ls U
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  - ch5_exec_examples U
  - ch6_pipe_sum U
  - ch7_pipeline U

Outline

Timeline

ch6_pipe_sum.c U | ch7_pipeline.c U | ch8_wait_nonblock.c U | ch9_zombie.c U

lab3-fork-exec > challenges > ch9_zombie.c

```
6   int main(void) {
25
26      printf("[parent] wait complete; zombie removed\n");
27
```

Problems | Output | Debug Console | Terminal | bash - lab3-fork-exec

```
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch8_wait_nonblock
Parent: child PID=36792 finished with status 1
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: child PID=36793 finished with status 2
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: child PID=36794 finished with status 3
Parent: all children finished
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ gcc -Wall -Wextra -std=gnu11 -g chal
lenges/ch9_zombie.c -o ch9_zombie
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch9_zombie
[parent] PID=38108 created child PID=38109
[parent] sleeping 10 seconds before wait()
[child] PID=38109 exiting immediately
[parent] wait complete; zombie removed
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ps -eo pid,ppid,state,cmd | grep ch9
_zombie
  38470    1609 S grep --color=auto ch9_zombie
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch9_zombie &
[1] 40385
[parent] PID=40385 created child PID=40391
[parent] sleeping 10 seconds before wait()
[child] PID=40391 exiting immediately
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ps -eo pid,ppid,state,cmd | grep '[c
]h9_zombie'
  40385    1609 S ./ch9_zombie
  40391   40385 Z [ch9_zombie] <defunct>
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ [parent] wait complete; zombie remov
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ps -eo pid,ppid,state,cmd | grep '[c
]h9_zombie'ppid,state,cmd | grep '[c]h9_zombie'
[1]+  Done                    ./ch9_zombie
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $
```

Chat

Build with Agent

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C ch9_zombie.c

Describe what to build

Agent | Models

Local | Default permissions

Explorer

linux-c-lab [Codespaces: urban space gig...

- c-programs
- command-line
- lab3-fork-exec
  - bin
  - challenges
    - ch1_single_fork.c U
    - ch2_multi_child.c U
    - ch3_exec_ls.c U
    - ch4_exec_worker.c U
    - ch5_exec_examples.c U
    - ch6_pipe_sum.c U
    - ch7_pipeline.c U
    - ch8_wait_nonblock.c U
    - ch9_zombie.c U
    - worker.c U
  - lab_report
  - src
    - exec_demo.c U
    - fork_demo.c U
    - pipe_demo.c U
    - pipeline_demo.c U
    - zombie_demo.c U
  - ch1_single_fork U
  - ch2_multi_child U
  - ch3_exec_ls U
  - ch4_exec_worker U
  - ch5_exec_examples U
  - ch6_pipe_sum U
  - ch7_pipeline U

Outline

Timeline

C ch6_pipe_sum.c U | C ch7_pipeline.c U | C ch8_wait_nonblock.c U | C ch9_zombie.c U

lab3-fork-exec > challenges > C ch9_zombie.c

```
 6  int main(void) {
25
26      printf("[parent] wait complete; zombie removed\n");
27
```

Problems | Output | Debug Console | Terminal | bash - lab3-fork-exec

```
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ls | grep "demo"
exec_demo
fork_demo
pipe_demo
pipeline_demo
zombie_demo
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ gcc -Wall -Wextra -std=gnu11 -g chal
lenges/ch8_wait_nonblock.c -o ch8_wait_nonblock
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch8_wait_nonblock
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: child PID=36792 finished with status 1
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
Parent: child PID=36793 finished with status 2
Parent: no child finished yet; continuing work...
Parent: no child finished yet; continuing work...
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@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ gcc -Wall -Wextra -std=gnu11 -g chal
lenges/ch9_zombie.c -o ch9_zombie
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch9_zombie
[parent] PID=38108 created child PID=38109
[parent] sleeping 10 seconds before wait()
[child] PID=38109 exiting immediately
[parent] wait complete; zombie removed
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ps -eo pid,ppid,state,cmd | grep ch9
_zombie
  38470    1609 S grep --color=auto ch9_zombie
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $
```

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C ch9_zombie.c

Describe what to build

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Explorer

linux-c-lab [Codespaces: urban space gig...

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- lab3-fork-exec
  - bin
  - challenges
    - ch1_single_fork.c U
    - ch2_multi_child.c U
    - ch3_exec_ls.c U
    - ch4_exec_worker.c U
    - ch5_exec_examples.c U
    - ch6_pipe_sum.c U
    - ch7_pipeline.c U
    - ch8_wait_nonblock.c U
    - ch9_zombie.c U
    - ch10_pool.c U
    - worker.c U
  - lab_report
  - src
    - exec_demo.c U
    - fork_demo.c U
    - pipe_demo.c U
    - pipeline_demo.c U
    - zombie_demo.c U
  - ch1_single_fork U
  - ch2_multi_child U
  - ch3_exec_ls U
  - ch4_exec_worker U
  - ch5_exec_examples U
  - ch6_pipe_sum U
  - ch7_pipeline U

Outline

Timeline

U | ch7_pipeline.c U | ch8_wait_nonblock.c U | ch9_zombie.c U | ch10_pool.c U

lab3-fork-exec > challenges > ch10_pool.c

```
6   int main(int argc, char *argv[]) {
72      return 0;
73  }
```

Problems | Output | Debug Console | Terminal | bash - lab3-fork-exec

```
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch9_zombie
[parent] wait complete; zombie removed
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ps -eo pid,ppid,state,cmd | grep ch9
_zombie
  38470    1609 S grep --color=auto ch9_zombie
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch9_zombie &
[1] 40385
[parent] PID=40385 created child PID=40391
[parent] sleeping 10 seconds before wait()
[child] PID=40391 exiting immediately
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ps -eo pid,ppid,state,cmd | grep '[c
]h9_zombie'
  40385    1609 S ./ch9_zombie
  40391   40385 Z [ch9_zombie] <defunct>
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ [parent] wait complete; zombie remov
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ps -eo pid,ppid,state,cmd | grep '[c
]h9_zombie'ppid,state,cmd | grep '[c]h9_zombie'
[1]+  Done                    ./ch9_zombie
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ gcc -Wall -Wextra -std=gnu11 -g chal
lenges/ch10_pool.c -o ch10_pool
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $ ./ch10_pool 3 file1 file2 file3 file
4 file5
[worker PID=43191] starting task: file3
[worker PID=43189] starting task: file1
[worker PID=43190] starting task: file2
[worker PID=43191] finished task: file3
[parent] worker PID=43191 finished
[worker PID=43210] starting task: file4
[worker PID=43189] finished task: file1
[worker PID=43190] finished task: file2
[parent] worker PID=43189 finished
[parent] worker PID=43190 finished
[worker PID=43211] starting task: file5
[worker PID=43210] finished task: file4
[parent] worker PID=43210 finished
[worker PID=43211] finished task: file5
[parent] worker PID=43211 finished
[parent] all tasks completed
@CssRyte ➜ /workspaces/linux-c-lab/lab3-fork-exec (lab3-fork-exec) $
```

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C ch10_pool.c

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